[Xe] $4f^0$ $5d^1$ $6s^2$ and it belongs to d block. Filling of 4f orbital starts from Cerium (Ce) and its electronic configuration is [Xe] $4f^2$ $5d^0$ $6s^2$. As we move from Cerium to other elements the additional electrons are progressively filled in 4f orbitals as shown in the table.

**Table : electronic configuration of Lanthanum and Lanthanoids**

| Name of the element | Atomic number | Symbol | Electronic configuration |
|---|---|---|---|
| Lanthanum | 57 | La | [Xe] $4f^0$ $5d^1$ $6s^2$ |
| Cerium | 58 | Ce | [Xe] $4f^2$ $5d^0$ $6s^2$ |
| Praseodymium | 59 | Pr | [Xe] $4f^3$ $5d^0$ $6s^2$ |
| Neodymium | 60 | Nd | [Xe] $4f^4$ $5d^0$ $6s^2$ |
| Promethium | 61 | Pm | [Xe] $4f^5$ $5d^0$ $6s^2$ |
| Samarium | 62 | Sm | [Xe] $4f^6$ $5d^0$ $6s^2$ |
| Europium | 63 | Eu | [Xe] $4f^7$ $5d^0$ $6s^2$ |
| Gadolinium | 64 | Gd | [Xe] $4f^7$ $5d^1$ $6s^2$ |
| Terbium | 65 | Tb | [Xe] $4f^9$ $5d^0$ $6s^2$ |
| Dysprosium | 66 | Dy | [Xe] $4f^{10}$ $5d^0$ $6s^2$ |
| Holmium | 67 | Ho | [Xe] $4f^{11}$ $5d^0$ $6s^2$ |
| Erbium | 68 | Er | [Xe] $4f^{12}$ $5d^0$ $6s^2$ |
| Thulium | 69 | Tm | [Xe] $4f^{13}$ $5d^0$ $6s^2$ |
| Ytterbium | 70 | Yb | [Xe] $4f^{14}$ $5d^0$ $6s^2$ |
| Lutetium | 71 | Lu | [Xe] $4f^{14}$ $5d^1$ $6s^2$ |

In Gadolinium (Gd) and Lutetium (Lu) the 4f orbital, are half-filled and completely filled, and one electron enters 5d orbitals. Hence the general electronic configuration of 4f series of elements can be written as [Xe] $4f^{2-14}$ $5d^{0-1}$ $6s^2$

## Oxidation state of lanthanoids:

The common oxidation state of lanthanoids is +3. In addition to that some of the lanthanoids also show either +2 or +4 oxidation states.

$Gd^{3+}$ and $Lu^{3+}$ ions have extra stability, it is due to the fact that they have exactly half filled and completely filled f-orbitals respectively.their electronic c onfigurations are

$$Gd^{3+} : [Xe]4f^7$$
$$Lu^{3+} : [Xe]4f^{14}$$



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